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PUBLIC SECTOR CAREERS

Hollywood, DARPA and innovation in the public service

FEATURE: Despite homogeneity, within government's human capital lie true entrepreneurs. DARPA's "special forces" model is an unconventional approach to unlocking potential, writes a veteran of wicked problems.



MARIE JOHNSON · FEBRUARY 19, 2016

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Last week, Peter Shergold delivered his manifesto for public sector transformation — calling for the "Hollywood model" of working, "where each film

project brings together
to read this.

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e". I urge everyone

Expectedly, there has been
Shergold's manifesto.

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a response to

Shergold wasn't criticising the dedicated efforts of public servants — or even calling for a different type of outsourcing. He was however, pointing out some inconvenient truths — that the Australian Public Service needs to open up and that adaptive government needs to be embraced.

But what does this really mean and how ready is the APS for this future characterised by openness, risk taking, innovation and adaptation.

The extent of the real challenge for adaptive government and innovation driven transformation is illustrated by the recent State of the Service Report and other APSC publications, which shows a rigid, aging and homogenous public service.

Those who are horrified at Shergold's manifesto, should be more horrified at the following statistics.

- 80% of APS level employees have only worked in one agency
- 36.6% of SES officers have only worked in one agency.
- Only 1.6% of APS employees moved between agencies
- The proportion of APS employees 50 years of age and over has grown significantly over time, increasing from 20.2% of all employees in June 2001 to 31.7% at June 2015.
- The 25–29 group decreased before levelling off at around 9%.

- The 20–24 group
- The average length
- There has only been 10% of people from non-English speaking countries

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down less than 2%.
the
people from non-

- There has been a decline in graduate numbers; and
- Only 34% of APS employees agree their agency applies merit appropriately.

Such homogeneity is not the fertile ground for innovation — which has always been driven by diversity, reinvention and the flow through of new perspectives to fuel ideas and immunize against risk aversion.

This is not a situation that can be quickly or easily reversed through traditional thinking or traditional and tired management strategies such as “outsourcing” and “shared services”.

However these statistics do not tell the full story — the numbers themselves present a narrow perspective with no analysis or future view, which is an issue in itself altogether.

Yes, the public service wants to be known for being innovative, and even digital. It talks about Uber and AirBnB and how good all this would be.

But it doesn't matter how many times the words “innovation” and its cousin “digital” are used in documents and organizational structures. The fact of the matter is that the human capital of the public sector has not been systematically cultivated and primed to unlock the full innovation potential of this pillar of Australian democracy — the Australian Public Service.

Now that's not to say that innovation doesn't happen in the APS — it certainly does, and there are some extraordinary examples of this. Public entrepreneurs are amongst us.

But here's the kick

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The response to ra

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Shergold's comm

ne as those to

riosity —

quirky. Definitely not systematic. Just different. Iconoclast.

As a consequence, broad generalizations fly “innovation does not happen in the public service” — often fuelled by media and arm chair critics and commentators sitting on the sides. There is an industry of commentators who have never actually had to implement anything.

Innovation has happened in spite of the system, and not because of it. To restate the situation: the level of homogeneity evidenced in the SoSR does not cultivate and prime for systemic innovation.

But if the system itself has not cultivated this innovation — then what has?

It is incredibly important — indeed essential — to understand this. As essential as it is to understand risk and failure as examined by Shergold, there is an equal imperative to understand innovation as an “outlier” in an homogenous system.

To understand the people, their way of thinking, their skills, experience, risk appetite, their networks, their relationships and teaming. It is not systematic or systemic — yet — but there are entrepreneurs working within and with the public sector who think and work differently. They come together to work on highly complex challenges within limited timeframes. And then move on. And they know the machinery of government.

These different thinkers are part of the public sector ecosystem — but not restricted to it.

This is Shergold's “Hollywood” approach — and the SoSR statistics do not tell this

story.

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The APSC in craftir

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system dynamic.

Over many years, I have spoken and written about a similar approach used by the Pentagon Defense Advanced Research Projects Agency (DARPA) – an approach akin to this I have used over time in the delivery of major programmes.

The DARPA model could well describe how Shergold's "Hollywood" might be systematically achieved.

Insight from DARPA

Back in October 2013, the Harvard Business Review carried an article called "Special Forces Innovation: How DARPA Attacks Problems".

This is a must read together with Shergold's manifesto. I urge all horrified and interested commentators to read both.

DARPA has a long track record of radical innovation. In my view, a radical innovation approach is necessary to break through many of the wicked problems confronting government administration. And wicked problems cannot be solved in a homogenous environment.

Among DARPA's phenomenal innovations, according to the HBR, are the Internet, global positioning satellites, stealth technology, and even micro-electrical-mechanical systems (MEMS) which are now pervasive in many products from air bags, ink-jet printers to video games like the Wii.

What struck me was the systematic approach DARPA takes to solving really complex problems — dealing with unknown and unknowable variables — and innovating along the way. And not with a big budget ...

According to the H
With an historic 50
group: "About 100
contract performe

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ee to five years.
eclectic
brant mix of
es, companies of

all sizes, labs, government partners and non-profits — to do the project work."

The DARPA approach has been described as unconventional but there is something in the organisation of the effort, the complexity of the challenge, and the temporary nature of diverse dynamic teams pursuing ambitious goals, that has proved enduring for DARPA.

What is particularly compelling in the DARPA approach is what the HBR describe as a "special forces" model of innovation. The DARPA "special forces" model of temporary targeted teams is compelling as a new approach to the innovation driven transformation of government over the next decade.

Two people who previously headed up DARPA and are now VPs at Google — Regina Dugan and Kaigham Gabriel — believe that the DARPA approach to breakthrough innovation is replicable. In my experience of this model, it is absolutely applicable to solving the wicked problems. Duggan and Gabriel speak about three essential elements.

1. *Ambitious goals.* The HBR article explains that "the problems must be sufficiently challenging that they cannot be solved without pushing or catalyzing the science." Further — and I find this a compelling insight into the power of ambitious targets — "that the presence of an urgent need for an application creates focus and inspires greater genius." "Catalyzing" and "genius" ... imagine these two powerful forces being brought to the challenge of the transformation of government and solving phenomenally complex problems.
2. *Temporary projects teams.* DARPA brings together "world class experts from industry and academia to work on projects of relatively short duration." These are not open-ended research programs. Dugan and Gabriel emphasise that the "intensity, sharp focus, and finite time frame make them attractive to the highest calibre talent, and the nature of the challenge inspires unusual levels of collaboration." Imagine unleashing this level of collaboration on the challenge of

complex societies
that in this way
with other great

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Gabriel point out
great problems

3. *Independence.*

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HBR article notes

projects and the
tion to move fast

and take bold risks and helps it persuade the best and brightest to join." The attribute of independence would also have application to the challenge of solving wicked problems. Dugan and Gabriel believe that such project groups function in ways that differ from the rest of the organisation; that breakthrough innovations may lead to major departures for how the organisation operates. I would agree with this — this is not business as usual. And nor does the attribute of independence diminish accountability.

The obvious point that Dugan and Gabriel make is that "when different outcomes are wanted, different approaches are necessary."

Radical innovation in action

The way I see it, Shergold's Hollywood approach and DARPA model have much in common, and I do believe that the DARPA model describes how Shergold's "Hollywood" could be systematically achieved.

The delivery of the technology enabling the National Disability Insurance Scheme and the cultivation of an innovation ecosystem is a current example of the DARPA approach in action.

This is a public purpose magnetizing diverse talents towards an immovable and compelling delivery date.

From a technology and innovation perspective, the best of the best from the public and private sectors. Disability entrepreneurs; Australian SMEs; different industries; global technology leaders; specialist individuals; research scientists; labs; people who run start-ups and ecommerce businesses in their "spare" time. This is a highly unusual, intensive and global level of collaboration that reaches to the Academy Award winner Dr Mark Sagar; global academic institutions, and the team of Sir Tim

Berners-Lee.

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This is in every way
is an ecosystem in

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del in action. This

A similar approach was used in the introduction of the New Tax System back in 2000.

The real disappointment with the State of the Service Report — beyond the picture of aging and homogeneity — is that it fails to envision the APS as part of this phenomenal ecosystem.

Shergold's manifesto lays down the challenge to systematize — and not bureaucratize — this model. The DARPA model describes how this could be achieved. We know the model works. The SoSR itself will need to adapt with an inspiring view of the future as the APS opens up. Innovation is not about counting people in silos.

About the author



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Marie Johnson is managing director and chief digital officer for the Centre for Digital Business. Previously, Marie has held senior executive roles in the public and private sectors in technology and innovation, and previously the Microsoft World Wide Executive Director Public Services and eGovernment.

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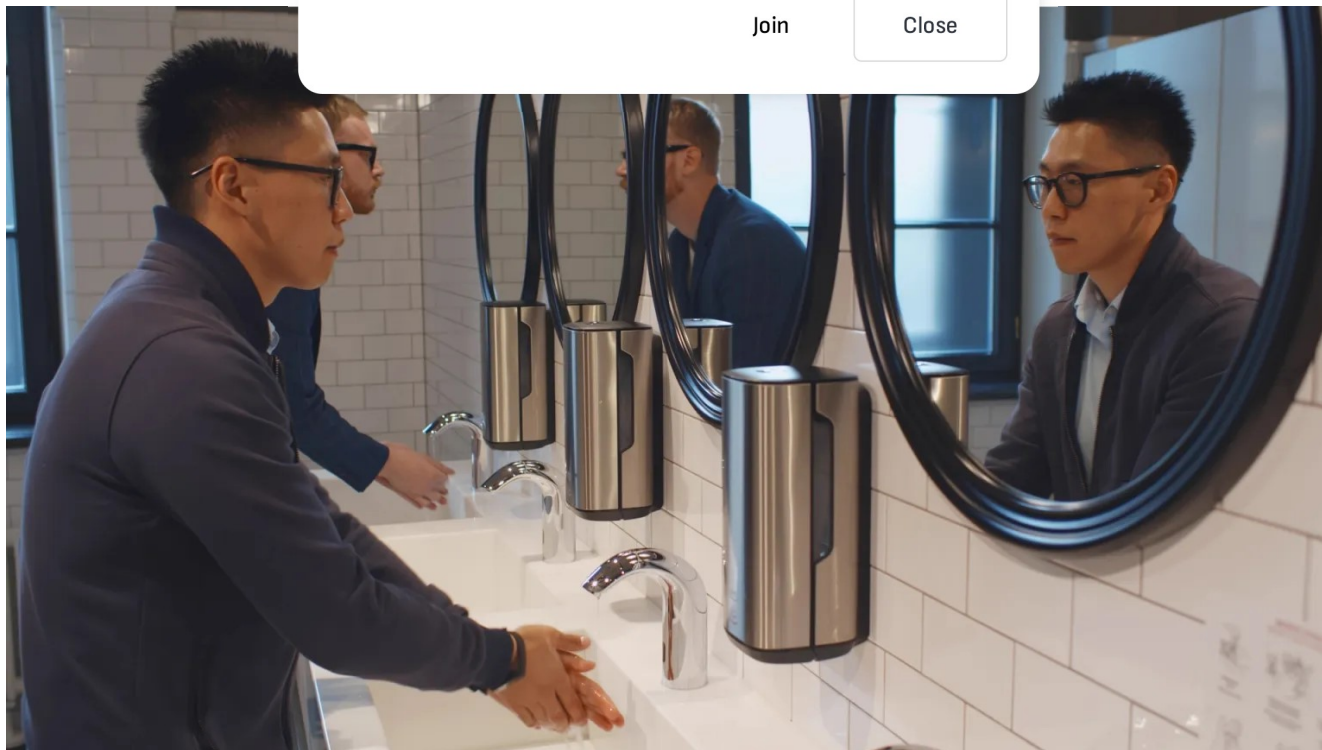
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